

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (III)	Branch: Computer Science & Engineering
Subject: Operating System	Course Code: B022315(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 3 T: 1 P: 0 Credits: 4
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

UNIT- I Introduction: Operation System objectives and functions, The Evolution of operating Systems, Batch Systems, interactive systems, time sharing and real time systems, Protection. Operating System Structure, System Components, operating system service, System structure. Distributed Computing.

UNIT-II Concurrent Processes: Process concept: Introduction, Definitions, Process States, Process State Transitions, The process Control Block, Operations on Processes, Suspend and Resume, Interrupt Processing. Mutual Exclusion, the Producer / Consumer problem, the critical section problem, Semaphores, Classical problems in concurrency, inter process communication. CPU scheduling: concepts, performance criteria, and scheduling Algorithms. Algorithm evaluation, Multiprocessor scheduling.

UNIT- III Dead Locks: System model, Deadlock characterization. Prevention, Avoidance and Detection, Recovery from deadlock, combined approach.

UNIT-IV Memory Management: Base machine, Resident Monitor, multiprogramming with fixed partition, Multiprogramming with variable partitions, Paging, Segmentation, paged - segmentation, virtual Memory concepts, Demand paging, performance, Page Replacement algorithms, Allocation of frames, Thrashing, cache memory organization.

UNIT-V I/O Management & Disk Scheduling: I/O system Interrupts Direct Memory Access, I/O Buffering, File system: File Concepts – File organization and Access mechanism, File Directories, File sharing, Implementation issues. Disk Scheduling algorithms. Case Study on LINUX: Kernel and Buffer Cache Architecture, concept of inode file & directory structure, Basic system calls (Open, Read, Write, namei, File and Record Close, File Creation, Creation of Special Files, Change Directory and Change Root, Change Owner and Change Mode

Text Books:

1. Operating system concepts Galvin by Silberschatz, John Wiley & Sons
2. Operating System Design & Implementation by Tanenbaum, A.S., PHI.
3. The Design of Unix Operating System, Maurice J. Bach, Pearson Education.

Reference Books :

1. Modern Operating System: Andrew S. Tanenbaum, PEARSON EDUCATION INTERNATIONAL
2. Operating System concepts by Silberschatz A and Peterson, J.L, PE- LPE.
3. Operating systems: Internals & Design Principles, William Stallings, PHI.

Course Outcomes [After undergoing the course, students will be able to:]

1. Identify the role of operating system in making computers execute data-processing jobs.
2. Realize managing computer's resource complexity during concurrent process execution through OS layers.
3. Analyse the reasons of resource bottlenecks-concurrency, deadlock and various synchronization mechanisms available.
4. Understand the functioning of operating system components in Memory Management techniques, Virtual Memory Management.
5. Understand disk organization, file system structure, Secondary Storage Management functions of OS.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (III)	Branch: Computer Science & Engineering
Subject: Operating Systems (UNIX) Laboratory	Course Code: B022323(022)
Total / Minimum-Pass Marks (End Semester Exam): 40 / 20	L: 0 T: 0 P: 2 Credit(s): 1

List of Experiments: (At least 10 experiments are to be performed by each student)

- Practice session: Study the features of Linux environment, basic Linux commands (echo, who, date, pwd, cd, mkdir, rmdir, ls, cp, mv, rm, cat, more, wc, find, tail, head, sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, comm, cmp, diff, tr); also document the syntax and semantics of those commands.
- Write a shell script that accepts a name from the user and displays whether it is a file, directory or something else.
- Write a shell script that creates users; also check if a particular user has logged in or not. If not, continue the loop till he/she logs in. Once the required user logs in, display a message.
- Write a shell script that searches for a given string in a text input file.
- Write a shell script that compiles all C files in your home directory and creates executable files.
- Write a shell script that given a filename as argument, deletes all even lines in a file & removes duplicate lines from a file.
- Write a shell script that enhances find command by adding error messages that explain why the command failed.
- Write a shell script to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer. Calculate percentage and grade according to following logic:
Percentage $\geq 90\%$: Grade A, Percentage $\geq 80\%$: Grade B, Percentage $\geq 70\%$: Grade C, Percentage $\geq 60\%$: Grade D, Percentage $\geq 40\%$: Grade E, Percentage $< 40\%$: Grade F.
- Write a shell script to accept the name, grade, and basic salary from the user. Write the details into a file called employee, separating the fields with a colon (,) continue the process till the user wants.
- Write an 'Awk' script to count the number of lines in a file that do not contain vowels.
- Write an 'Awk' script to find the number of characters, words and lines in a file.
- Write a C program to simulate following non-preemptive CPU scheduling algorithms: a) FCFS b) SJF c) Round Robin (pre-emptive) d) Priority scheduling techniques.
 - to find average turnaround times and waiting times;
 - to display / print Gantt Chart (in any convenient format).
- Implement the Producer – Consumer problem using semaphores (using UNIX system calls).
- Write a C program to simulate disk-scheduling algorithms: a) FCFS b) SCAN c) C-SCAN techniques.
- Write a C program to simulate page replacement algorithms: a) FIFO b) LRU c) LFU d) OPT techniques.

Remarks: The laboratory experiments may be performed in with of the LINUX shell environments: BOURNE Shell / KORN Shell / 'C' Shell.

Laboratory Outcomes [After undergoing the course, students will be able to:]

- Understand the concept of Unix and shell programming.
- Learn the working of Linux OS Kernel.
- Analyse the differences between features provided in Windows and Linux operating system.
- Learn the concept of loops and decision-making statements.
- Analyse the logic & procedure of problem solving through Scripts.

Recommended Books:

- Advance UNIX, a Programmer's Guide, S. Prata, BPB Publications, New Delhi.
- The Complete Reference Unix, Rosen, Host, Klee, Farber, Rosinski, Second Edition, TMH.

Printed Pages – 5

Roll No. :

B022315(022)

**B. Tech. (Third Semester) Examination,
Nov.-Dec. 2021**

AICTE (New Scheme)

(CSE Branch)

OPERATING SYSTEMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks 35

Note : Attempt all questions. Part (a) of each question is compulsory. Attempt any two from (b), (c) and (d) parts of each question. Part (a) carry 4 marks each and part (b), (c) and (d) carry 8 marks each.

Unit-I

1. (a) What do you mean by time sharing operating? Give an example.

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- (b) What are the functions of operating system? Explain in brief. 8
- (c) Explain the working and architecture of an operating system in detail. 8
- (d) Explain the various system calls in an operating system. 8

Unit-II

2. (a) Define process control block with proper diagram. 4
- (b) Consider the above set of processes that arrive at time zero. Calculate the average waiting time, average turnaround time and throughput with FCFS and SJF. 8

Process	Burst time
P ₁	5
P ₂	24
P ₃	16
P ₄	10
P ₅	3

[3]

- (c) What do you mean by Dining Philosopher method? Explain with suitable diagram. 8
- (d) Explain Producer-Consumer problem in detail. 8

Unit-III

3. (a) Define resource allocation graph with suitable diagram. 4
- (b) Explain the method used in deadlock prevention. 8
- (c) What are the four necessary conditions required to hold a deadlock? 8
- (d) An operating system uses the banker's algorithm for deadlock avoidance when managing the allocation of three resource types X, Y, and Z to three processes P0, P1 and P2. The table given below presents the current system state. Here, the allocation matrix shows the current number of resources of each type allocated to each process and the max matrix shows the maximum number of resources of each type required by each process during its execution. 8

[4]

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	0	6	2	0
P2	2	1	1	3	3	3

There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in safe state. Consider the following independent requests for additional resources in the current state :

REQ 1 : P0 requests 0 units of X, 0 units of Y and 2 units of Z

REQ 2 : P1 requests 2 units of X, 0 units of Y and 0 units of Z

Unit-IV

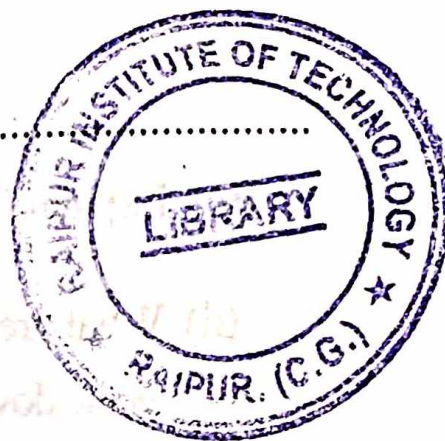
4. (a) Define resident monitor. 4
- (b) Explain address translation from logical address to physical address. 8

[5]

- (c) What is page replacement algorithm? Explain any one algorithm with example. 8
- (d) What is thrashing? State the cause of thrashing. 8

Unit-V

5. (a) What is buffering? 4
- (b) Explain virtual machine operating system. 8
- (c) Explain file organization and access mechanism. 8
- (d) Explain the various disk scheduling algorithm with suitable example. 8



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B. Tech. (Third Semester) Examination,

Nov.-Dec. 2022

(CSE Branch)

(New Scheme)

OPERATING SYSTEMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) of each question is compulsory & carry equal 4 marks. Attempt any two from (b), (c) and (d) & carry equal 8 marks.

Unit-I

1. (a) What are the objective of Operating System.
- (b) Differentiate between time sharing system and Real time system.

(c) Explain Distributed Computing.

(d) What are the problem in Batch Operating System?
Write down their solution with the proper explanation.

Unit-II

2. (a) What is Process. Draw & explain the process state, diagram.

(b) Write short notes on :

- (i) The process control block
- (ii) The critical section problem

(c) Define Semaphore? Give the solution of producer consumer problem using semaphore.

(d) Consider the set of 5 process whose arrival time & burst time are given below :

Process No.	Arrival time	Burst time
P ₁	0	3
P ₂	2	6
P ₃	4	4
P ₄	6	5
P ₅	8	2

Find :

- (i) Average waiting time
- (ii) Turn around time
- (iii) Average turn around time
- (iv) Through put

for shortest remaining time first algorithm.

Unit-III

3. (a) What is safe and unsafe state.

(b) Explain the deadlock recovery method.

(c) Explain how the deadlock can be prevented.

(d) A single processor system has 3 resource types x, y, z which are shared by 3 processes. There are 5 units of each resource type. Consider the following scenario, where the column allocation denotes the number of units of each resource type allocated to each process and the column request denotes the number of units of each resource type requested by a process in order to complete execution. Which of these processes will last finish and are the system is in safe state.

[4]

	Allocation			Request		
	x	y	z	x	y	z
P ₀	1	2	1	1	0	3
P ₁	2	0	1	0	1	2
P ₂	2	2	1	1	2	0

Unit-IV

4. (a) What is virtual memory?
- (b) Explain the address translation from logical to physical address.
- (c) Write short notes on :
- Paging
 - Segmentation
- (d) Consider a reference string 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. the number of frames in the memory is 3 find out the number of page faults respective to (i) optimal (ii) FIFO (iii) LRU.

Unit-V

5. (a) What is DMA.

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[5]

- (b) Differentiate between program driven and interrupt driven input/output.
- (c) Explain Kernel and Buffer cache architecture of Unix Operating System.
- (d) Consider a disk with 200 tracks and the queue has random requests from different processes in the order :
- 55, 58, 38, 18, 90, 160, 150, 38, 184
- initially arm is at 100. Find the average seek length using FIFO, SSTF, Scan and C-scan algorithm.

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B. Tech. (Third Semester) Examination,

Nov.-Dec. 2020

(CSE Branch)

(Specialization : CSE)

OPERATING SYSTEMS

Time Allowed : Three hours

Maximum Marks : 100

Note : Attempt all questions. Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d).

Unit-I

1. (a) Explain how operating system acts as a Resource Manager?

4

- (b) The services and functions provided by an operating

[2]

system can be divided into different components. Briefly explain all the components of Operating System.

8

- (c) Analyze different types of operating systems with an example. Which type of operating system is most suitable for the modern computer system and why? 8
- (d) Name and discuss the range of system calls that may be supplied by an operating system under various circumstances. List the sequence of system call required to copy content of a file to another file. 8

Unit-II

2. (a) Give Five State model of process state transition. 4
- (b) What are the two models of inter-process communication? What are the strengths and weakness of the two approaches? 8
- (c) Discuss critical section problem and how this can be overcome using semaphores. Write the solution for achieving mutual exclusion requirement in a producer consumer problem. 8

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- (d) What is CPU scheduling? If the different jobs and their arrival time and burst time are given below :

P. No.	Arrival time	Burst time
1	0	6
2	1	4
3	2	3
4	3	1
5	4	2
6	5	1

Find the average turnaround time, average waiting time and throughput using Round Robin (with time quantum = 2) and SRTF. 8

Unit-III

3. (a) What is Deadlock? Give characteristics of deadlock. 4
- (b) What are the four necessary conditions for characterizing deadlock? What is the only reasonable condition that can be used to prevent deadlocks from occurring? 8
- (c) Explain Banker's algorithm and safety algorithm with pseudo code. 8

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- (d) Write an algorithm to determine whether a given system is in a deadlock and explain. Consider resource A has total 7 instances, B has 2 instances and C has 6 instances the following snapshot of a system :

Process	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	0	0	0	0	0	0
P ₁	2	0	0	2	0	2			
P ₂	3	0	3	0	0	0			
P ₃	2	1	1	1	0	0			
P ₄	0	0	2	0	0	2			

Answer the following using Banker's algorithm :

- Is the system in a safe state?
- If Yes, mention the safe sequence.
- If a process P₂ makes a request of one additional request for an instance of type C, will the system lead to deadlock state then.

8

Unit-IV

4. (a) What is virtual memory?

4

[5]

- (b) What is the need of page replacement? Consider the following page references string :
1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.
Find out the no. of page fault by LRU, OPT replacement, FIFO algorithm. Assume there are four page frames.

8

- (c) Explain the difference between internal and external fragmentation :

Given five variable sized memory partitions of 100 kb, 500 kb, 200 kb, 300 kb, 600 kb (in order), how would each of the first fit, best fit and worst fit algorithm place processes of 212 kb, 417 kb, 112 kb and 426 kb (in order)? Which algorithm makes most efficient use of memory?

8

- (d) Why are segmentation and paging sometimes combined into one scheme? Explain the paged segmentation.

8

Unit-V

5. (a) What are the different files allocation methods?
- (b) Considering SCAN disk scheduling algorithm. How

4

is it possible to reduce the time required to finish off?
Explain with an example. 8

(c) Describe various file accessing methods with its advantages and disadvantages. 8

(d) Suppose that the head of a moving head disk with 200 tracks, 0 to 199, is currently serving a request at 143 and has just finished a request at track 125. The queue of request is kept in FIFO order 86, 147, 91, 177, 94 150, 102, 175, 130. What is the total number of head movement needed to specify these request for the following disk scheduling algorithms?

(i) SSTF scheduling

(ii) SCAN scheduling 8

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**B. Tech. (Third Semester) Examination,
April-May 2021**

(AICTE Scheme)

(CSE Branch)

OPERATING SYSTEMS

Time Allowed : Three hours

Maximum Marks : 100

Passing marks - 35

Note : Attempt all questions. Part (a) of each question is compulsory. Attempt any two from (b), (c) and (d) parts of each question. Part (a) carry 4 marks each and part (b), (c) and (d) carry 8 marks each.

Unit-I

1. (a) What are the 3 main purposes of an Operating System?

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[2]

- (b) Compare tightly coupled systems with loosely coupled systems. 8
- (c) Write a brief note on different operating system structures. 8
- (d) Distinguish among the following terminologies associated with the operating system and explain each of them in detail. 8
- Multiprogramming systems, Multitasking systems, Multiprocessor systems.

Unit-II

2. (a) What do you mean by PCB? Where is it used? What are its contents? Explain. 4
- (b) Explain Short Term, Long Term and Medium Term Scheduler. 8
- (c) What are the different states of process explain each of them briefly. 8
- (d) Consider the following 4 processes with length of cpu burst time given in milliseconds :

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Process	Burst time	Arrival time
P1	1	3
P2	4	1
P3	2	4
P4	6	0
P5	3	2

Draw Gantt's chart and find out the average waiting time for SJF processor scheduling algorithm. 8

Unit-III

3. (a) What is Deadlock? Explain the necessary conditions for its occurrence. 4
- (b) Explain the resource allocation graph. 8
- (c) Write down the different techniques which we can incorporate to prevent deadlock. 8
- (d) Why is deadlock state more critical than starvation? Describe resource allocation graph with a deadlock, with a cycle but no deadlock. 8

Unit-IV

4. (a) What is Paging and Swapping? 4

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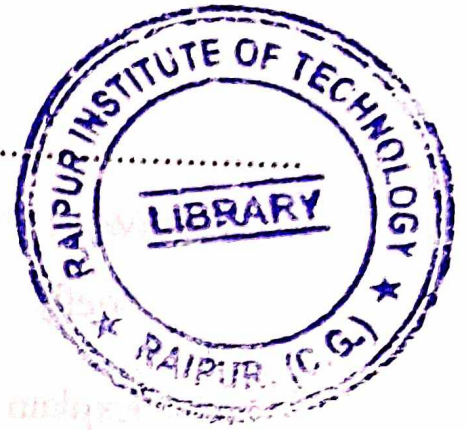
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[4]

- (b) Define external fragmentation. What are the causes for external fragmentation? 8
- (c) Explain demand paging with an example. 8
- (d) What is Thrashing? What are the causes for thrashing? 8

Unit-V

5. (a) Define logical and physical address space. 4
- (b) Explain the following : 8
- (i) file types
 - (ii) file operation
 - (iii) file attributes
- (c) What is Virtual Memory? Explain overlays method with the help of example: 8
- (d) Suppose that the Request sequence = {176, 79, 34, 60, 92, 11, 41, 114}
Initial head position = 50
Direction = left (We are moving from right to left)
Find out the total seek count using SCAN disk scheduling algorithm. 8



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**B. Tech. (Third Semester) Examination,
April-May 2024**

(New Scheme)

(CSE Branch)

OPERATING SYSTEM

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part(a) of each question in compulsory carrying 4 marks. Attempt any two parts from parts (b), (c) and (d) carrying 8 marks each.

1. (a) Define Kernel.
- (b) Explain Operating system structure with suitable diagram.

- (c) What are the services of Operating System? Explain briefly.
- (d) Explain time sharing and real time operating system.
2. (a) Define Race condition with example.
- (b) Write the difference between process and thread.
- (c) Explain the concept of Dining Philosopher problem with suitable diagram and representation.
- (d) Consider a system with set of processes with burst time (in ms) and priorities (consider 1 to be highest). Draw Gantt chart and find Average waiting time for pre-emptive priority scheduling.
- | Process | Burst time | Priority | Arrival Time |
|---------|------------|----------|--------------|
| P1 | 10 | 3 | 0 |
| P2 | 5 | 2 | 1 |
| P3 | 2 | 1 | 2 |
3. (a) Define Deadlock.
- (b) Explain how deadlock can be prevented on the basis of all 4 deadlock conditions.

- (c) Consider the following system with processes from P_0 to P_4 .

Process	Allocation	Max	Available
	A B C	A B C	A B C
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

Using Bankers algorithm find the following :

- (i) Content of Need Matrix
- (ii) Is the system in a safe state? If yes then what is the safe sequence.
- (d) Write short notes on :
- (i) Resource Allocation Graph
- (ii) Recovery from deadlock
4. (a) Define Logical and Physical Address.
- (b) Explain the concept of Segmentation with suitable diagrams.

- (c) Determine the number of page fault having 3 frames available in memory, when reference to pages occur in order :

4,7,6,1,7,6,1,2,7,2

(i) OPR

(ii) LRU

- (d) What is virtual memory? How it implemented using demand paging explain?

5. (a) Define File and its attributes.

- (b) Describe File Access methods in detail.

- (c) Suppose the moving head disk with 200 tracks is currently serving a request for track 143. If the queue request is kept in FIFO order :
86,147,91,177,94,150;

What is the total head movement for the following scheduling schemes :

(i) FCFS

(ii) SSTF

- (d) What is inode? Write about *namei* algorithm.

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Roll No. :

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**B. Tech. (Third Semester) Examination,
April-May 2022**

(AICTE Scheme)

(Computer Science Engg. Branch)

OPERATING SYSTEMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks: 35

Note : Attempt all questions. Part (a) of each question is compulsory. Attempt any two from (b), (c) and (d) parts of each question. Part (a) carry 4 marks each and part (b), (c) and (d) carry 8 marks each.

Unit-I

1. (a) Explain how an operating system tasks control?
- (b) Explain operating system structure.

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- (c) What the various operating system services?
- (d) Define multiprogramming and real time operating system.

Unit-II

2. (a) Demonstrate race condition.
- (b) Describe the objectives of Scheduling. Explain the structure of PCB.
- (c) Explain semaphore. How can semaphore be used to Enforce mutual – exclusion? Give a suitable example to explain.
- (d) Consider the following set of processes having their burst in ms. calculate the avg waiting time and turn-around time for SJF (non preemptive)

Process	Burst time	Arrival time
P1	10	0
P2	1	1
P3	2	2
P4	1	3
P5	5	4

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[3]

Unit-III

3. (a) What is Deadlock? Draw suitable diagram.
- (b) Explain dead lock avoidance algorithm.
- (c) Analyze the following snapshot of a system.

	ALLOCATION				MAXIMUM				AVAILABLE			
	A	B	C		A	B	C		A	B	C	
P0	0	1	0		7	5	3		2	3	0	
P1	2	0	0		3	2	2					
P2	3	0	2		9	0	2					
P3	2	1	1		2	2	2					
P4	0	0	2		4	3	3					

- (i) Calculate the content of need matrix.
- (ii) Is the system in a safe state?
- (d) Explain Deadlock prevention in detail.

Unit-IV

4. (a) Define Virtual Address.
- (b) Explain fragmentation with its types and solution.
- (c) Consider the following page reference string
- 1, 1, 2, 2, 1, 4, 2, 3, 3, 5, 5, 4

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Assume 3 page frame, calculate the page faults using following algorithm.

- (i) FIFO
 - (ii) OPR
- (d) Explain the concept of paging and segmentation.

Unit-V

5. (a) What is Address Binding?
- (b) On a disk with 200 cylinders, compute the number of tracks the disk arm must move to satisfy all the request in the disk queue. Assume the current position is at 53 track are : 98, 183, 37, 122, 14, 124, 65, 67. Identify the total head movement for the following algorithm.
- (i) FIFO
 - (ii) SSTF
- (c) Describe file allocation methods.
- (d) Write in detail about File Access Mechanism.

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**B. Tech. (Third Semester) Examination,
Nov.-Dec. 2024**

(CSE Branch)

(New Scheme)

OPERATING SYSTEMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) of each question is compulsory & carries equal 4 marks each. Attempt any two from (b), (c) and (d) & carries equal 8 marks each.

Unit-I

1. (a) Explain how an operating system tasks control? 4
- (b) Explain operating system structure. 8

- (e) What are the various operating system services? 8
- (d) Define multiprogramming and real time operating system. 8

Unit-II

2. (a) Demonstrate race condition. 4
- (b) Describe the objectives of scheduling. Explain the structure of PCB. 8
- (c) Explain semaphore. How can semaphore be used to enforce mutual-exclusion? Give suitable example to explain. 8
- (d) Consider the following set of processes having their burst in ms. Calculate the avg waiting time and turnaround time for SJF (non preemptive).

Process	Burst time	Arrival time
P ₁	1	3
P ₂	4	1
P ₃	2	4
P ₄	6	0
P ₅	3	2

Unit-III

3. (a) Writing necessary conditions for deadlock. 4

(b) Consider the following snapshot of a system : 8

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	7	5	3	3	3	2
P ₁	2	0	0	3	2	2			
P ₂	3	0	2	9	0	2			
P ₃	2	1	1	2	2	2			
P ₄	0	0	2	4	3	3			

Answer the following using Banker's algorithm.

(i) What is the content of matrix needed?

(ii) Is the system in a safe state?

(c) Explain the deadlock recovery method. 8

(d) Write short notes on : 8

(i) Circular wait

(ii) Safe and unsafe state

Unit-IV

4. (a) What is Resident Monitor? 4
- (b) Explain multiprogramming with fixed partition with suitable example. 8
- ~~(c)~~ Consider the following page reference string : 8
- 4, 7, 6, 1, 7, 6, 1, 2, 7, 2
- Assume 3 page frames and pure demand paging.
- How many page faults would occur for :
- (i) FIFO
- (ii) Optimal
- (c) What is Thrashing? How does the system detect thrashing? 8

Unit-V

5. (a) Explain blocked I/O devices. 4
- (b) Explain the different types of buffering. 8
- (c) Write different types of file access methods with suitable diagram. 8
- (d) Write short notes on : 8
- (i) File sharing
- (ii) Virtual O.S.



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**B. Tech. (Third Semester) Examination,
April-May 2023**

(CSE Branch)

(New Scheme)

OPERATING SYSTEMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) of each question is compulsory & carry equal 4 marks. Attempt any two from (b), (c) and (d) & carry equal 8 marks. State assumptions clearly (if any).

Unit-I

1. (a) What is an operating system, explain.
- (b) Briefly describe the various services of an operating system.
- (c) Describe the main developments of operating system that occurred at each computer generation.

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- (d) What do you mean by system call? Explain with example.

Unit-II

2. (a) What is difference between Independent process and dependent process?
- (b) Explain various states of a process with state transition diagram.
- (c) Explain process control block.
- (d) If the different jobs and their arrival time and burst time is given below find average waiting time using FCFS & preemptive SJF (SRTF).

Process No.	Arrival time	Burst time
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

Unit-III

3. (a) Define deadlock with example.
- (b) How will you avoid deadlock?

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- (c) Explain Banker's algorithm & Safety algorithm.
- (d) Consider following current allocation :

Process	Allocation	Max	Available
	R ₁ R ₂ R ₃	R ₁ R ₂ R ₃	R ₁ R ₂ R ₃
P ₁	2 2 3	3 6 8	7 7 10
P ₂	2 0 3	4 3 3	
P ₃	1 2 4	3 4 4	

- (i) Is the current allocation state safe
- (ii) Would the following request be granted

In the current state

Process P₁ request (1, 1, 0)

Process P₂ request (0, 1, 0)

Unit-IV

4. (a) Explain the main function of the memory management unit.
- (b) Explain the concept of thrashing with example.
- (c) Consider a swapping system in which memory consists of the following hole size in memory order 10 kB, 4 kB, 20 kB, 18 kB, 7 kB, 9 kB, 12 kB and 15 kB. Which hole is taken for successive segment requests of :

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PTO

- (i) 12 kB
- (ii) 10 kB
- (iii) 9 kB

For first fit, best fit?

- (d) Explain paging with example.

Unit-V

5. (a) Explain the concept of Inode.

- (b) Write different file attributes and operations.

- (c) Explain the namei algorithm with the help of example.

- (d) Suppose that a disk drive has 181 cylinders, numbered 0 to 180. The drive is currently serving a request at cylinder 60, and the previous request was at cylinder 50. The queue of pending request, in FIFO order is 87, 170, 40, 150, 36, 72, 66, 15. Starting from the head position, what is the average head movement and what is the total no. of head movement needed to satisfy these request for the following disk scheduling algorithms

- (i) FCFS
- (ii) SSTF
- (iii) Scan



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**B. Tech. (Third Semester) Examination,
Nov.-Dec. 2023**

(CSE Branch)

(New Scheme)

OPERATING SYSTEMS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) of each question is compulsory & carry equal 4 marks. Attempt any two from (b), (c) and (d) & carry equal 8 marks.

Unit-I

1. (a) What is an Operating System? What it does as a *Resource Manager*?
- (b) Briefly explain any 5 services of Operating System, how it provides convenience to the user.

(c) What are the different types of operating system?

(d) Explain Layered Approach of Operating System.

Unit-II

2. (a) Draw and briefly explain Process State Transition diagram.

(b) Explain Dining philosopher problem with its possible solution.

(c) Give a mutual - exclusion solution for Producer-consumer problem of critical section with algorithm and example.

(d) What is CPU scheduling? If the different jobs and their arrival time and burst time is given below find the average turnaround time and average waiting time using FCFS and SJF.

Process No.	Arrival time	Burst time
0	0	4
1	1	5
2	2	2
3	3	3
4	4	6

Unit-III

3. (a) What is deadlock? Give any real life example of deadlock.

(b) What are the four necessary conditions for deadlock occurrence? Explain.

(c) Explain various methods of avoid deadlock with example.

(d) Determine whether a given system is in a deadlock. Consider the following snapshot of a system :

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	7	5	3	3	3	2
P ₁	2	0	0	3	2	2			
P ₂	3	0	2	9	0	2			
P ₃	2	1	1	2	2	2			
P ₄	0	0	2	4	3	3			

Answer the following using Banker's algorithm.

(i) What would be the need matrix?

(ii) Is the system in a safe state?

- (iii) If a request from process P1 arrives for (1,0,2) can be granted immediately?
- (iv) Explain various methods for recovery from deadlock.

Unit-IV

- 4. (a) Why do we need paging?
- (b) What is the need of Page replacement? Explain any one page replacement algorithm with suitable example.
- (c) What is Thrashing? State the cause of thrashing.
- (c) Write short notes on :
 - (i) Virtual memory
 - (ii) Cache memory

Unit-V

- 5. (a) What are the different operations performed on file?
- (b) Explain free space management and its techniques.
- (c) Describe various file accessing methods with its advantages and disadvantages.
- (d) What do you mean by Input/Output Buffering? Explain with their types.